SPMM Smart Revise

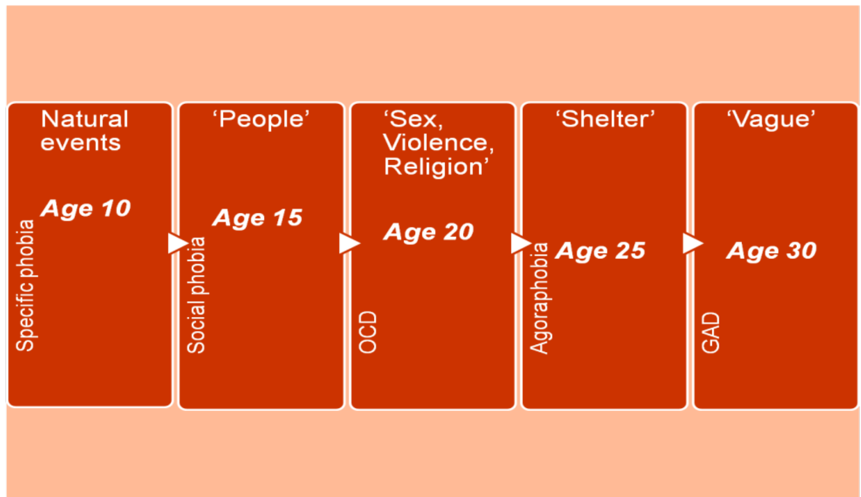
Adult Psychiatry II

Paper B **Syllabic content 7.1x**

1. Anxiety Disorders - Overview

15% of adults in the UK at any time have **broadly defined neurosis** according to National Psychiatric Morbidity Survey (NPMS, 1995). In this study, the UK prevalence of anxiety disorder was higher than depression (consistent with ECA and NCS studies from the USA). Phobic disorder the most common in ECA & NCS but in NPMS – the most common anxiety disorder **was mixed anxiety-depression** followed by GAD. 25% of all GP consultations are for anxiety related symptoms.

Considering the mean age of onset for various anxiety disorders, the following pattern is noted; GAD – 30 years, Panic disorder – 22 to 25 years, OCD – 20 years, social phobia – 15 years, specific phobia – varies with individual categories – blood injury injection or environmental types start around age 5 to 9 years while situational phobia starts around 20 years of age.



It is noted that **with each generation**, anxiety disorders are diagnosed at a younger age than the previous cohort.

The estimated lifetime prevalence of **blood-injection-injury phobia** is around 3.5%. The median age of onset is around 5 to 6 years. Subjects with blood-injection-injury phobia have higher lifetime histories of fainting and seizures. Prevalence was lower in the elderly and higher in females and persons with less education.

Sex distribution of anxiety disorders is an important topic often tested in the exams. As a general rule, all anxiety disorders are common in women than men. Notable exceptions are OCD, which is more in boys than girls, but equally common in adult men and women. Also, men outnumber women in attending health care centres for social phobia.

Summary of NICE guidance: treatment for generalized anxiety disorder, panic disorder and OCD
➢ A 'stepped care' approach to help in choosing the most effective intervention ➢ A comprehensive assessment that considers the degree of distress and functional impairment; the effect of any co-morbid mental illness, substance misuse or medical condition; and past response to treatment ➢ Treat the primary disorder first ➢ Psychological therapy is more effective than pharmacological therapy and should be used as first line where possible. ➢ Pharmacological therapy is also effective. Most evidence supports the use of SSRIs ➢ Consider combination therapy for complex anxiety disorders that are refractory to treatment

2. Obsessive Compulsive Disorder:

Epidemiology: The point prevalence of OCD is as high as **1-3% of adults** and 1-2% of children and adolescents (community samples). The lifetime prevalence of OCD is fairly constant – around 2-3%. It is the fourth most commonly prevalent psychiatric disorder in epidemiological studies (after phobias, alcohol use, and depression). The gender ratio of OCD in clinical samples is roughly equal, but females predominate in community populations (~1.5:1), this may be because men have more severe psychopathology. Males with OCD showed an earlier onset and a trend toward more tics and poorer outcome than females. (Heyman et al., 2006)

Presentation: Recent factor analytic studies of obsessions and compulsions in OCD tend to show a four-factor model (Castle & Phillips, 2006):

- Aggressive, sexual, and religious obsessions, and checking compulsions;
- Symmetry and ordering obsessions and compulsions; (often chronic and treatment resistant)
- Contamination obsessions and cleaning compulsions;
- Hoarding obsessions and compulsions (may be neurobiologically distinct and often treatment resistant)

Aetiology: The aetiology of OCD is unknown. Hypermetabolism of basal ganglia structures (caudate) has been reported in several neuroimaging studies. D2 antagonists such as clozapine and other antipsychotics can cause secondary OCD-like symptoms in some patients. Children with autoimmune reactions (e.g. PANDAS -see below) have a higher prevalence of OCD. OCD also overlaps with a spectrum of other disorders. OCD spectrum disorders can be classified as:

- 1. Those associated with ***somatic preoccupation*** e.g.body dysmorphic disorder, hypochondriasis or anorexia nervosa.
- 2. Those associated with ***neurological disorders*** (repetitive behaviours) e.g. Tourette syndrome, Sydenham's chorea and autistic spectrum.
- 3. Those associated with ***impulse control disorders*** or with rousing or pleasurable repetitive behaviours e.g. paraphilias, kleptomania, trichotillomania, and pathological gambling. (Castle & Phillips, 2006)
- Anankastic personality can also be considered a part of OCD spectrum.

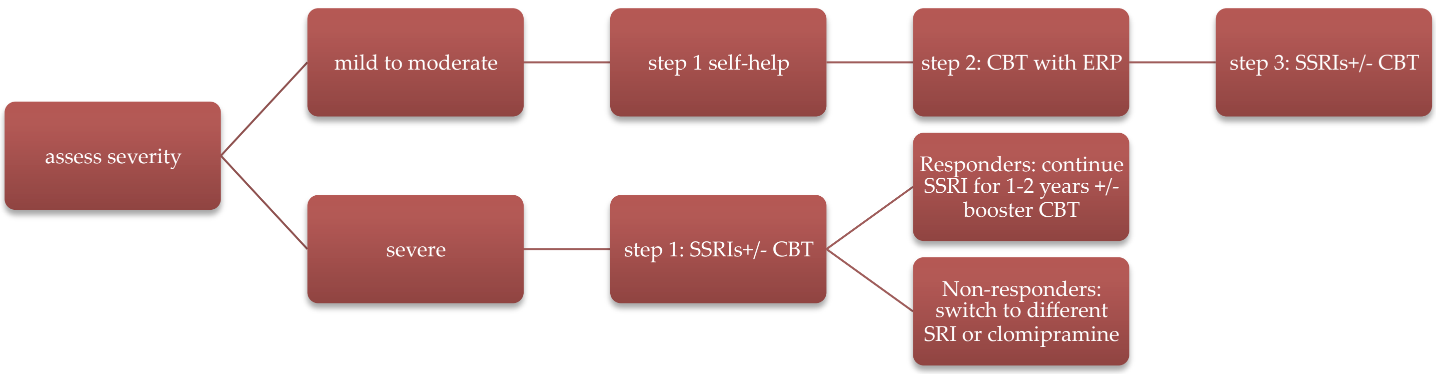
PANDAS: Paediatric autoimmune neuropsychiatric disorders associated with streptococcal infection—PANDAS is thought to be secondary to streptococcal infection and mediated by autoantibodies binding to basal ganglia. This produces tics, fluctuating obsessive-compulsive symptoms and anxiety.

National Institute of Mental Health Clinical Diagnostic Criteria for PANDAS
1. The presence of OCD or a tic disorder.
2. Onset between 3 years of age and the beginning of puberty.
3. Abrupt onset of symptoms or a course characterized by dramatic exacerbations of symptoms.
4. The onset or the exacerbations of symptoms is temporally related to infection with GABHS.
5. Abnormal results of the neurologic examination (hyperactivity, choreiform movements, and/or tics) during an exacerbation.

Note that the neuropsychiatric symptoms need not have onset during the **streptococcal (group A beta-hemolytic) infection**; exacerbation correlated temporally is also acceptable for a diagnosis.

AntiDNAseB or Antistreptolysin O titres are likely to be elevated in most with recent streptococcal infection. A fraction of these may have autoantibodies to neurons in basal ganglia called anti basal ganglia antibodies.

Treatment of OCD (according to NICE guidelines)



Exposure and response prevention is a key element of CBT for of OCD. Patients are trained to confront (directly or in imagination) anxiety-provoking situations while abstaining from compulsive behaviours in response. Periodic exposure and response prevention "booster" sessions can reduce the risk of relapse and provide more durable remission than using pharmacotherapy alone.

OCD shows a selective response to **serotonergic medications** such as clomipramine and SSRIs. Around 60 to 70% of patients experience some degree of improvement after SSRI treatment (typically at a higher dose, given for a longer duration than for depression). NNT for SSRIs (vs. placebo) varies between 6 and 12. **Antipsychotic augmentation** is indicated if no response is seen after a three-month trial of a maximal dose of SSRI. This is especially useful when tics are also seen.

3. Post-Traumatic Stress Disorder:

Diagnostic criteria for PTSD have been altered in DSM-5. PTSD (as well as Acute Stress Disorder) have been moved from anxiety disorders into a new class of "trauma and stressor-related disorders." A history of exposure to a traumatic event that meets specific stipulations and symptoms from each of four symptom clusters: intrusion, avoidance, negative alterations in cognitions and mood (including self-blame or blaming others persistently), and alterations in arousal and reactivity (including reckless and destructive behaviour). A clinical subtype "with dissociative symptoms" has also been added.

Epidemiology: The **incidence** varies across the world. Point prevalence is 1%. The National Comorbidity Survey Replication (NCS-R) found the lifetime prevalence of PTSD among adult Americans to be 6.8%. The **lifetime prevalence** among men was 3.6%, and among women was 9.7%. It is unclear whether the **gender difference** is due to higher exposure to trauma or greater vulnerability to develop PTSD.

Males are more likely than females to be exposed to traumatic events (60% males vs. 50% females). But females develop PTSD two times more frequently (12% vs. 6% in NCS) even after controlling the type of trauma. It is unclear if this is due to higher exposure to trauma or greater vulnerability to develop PTSD.

Up to 30% of people exposed to trauma may develop PTSD. The most frequently experienced trauma is witnessing someone being badly injured or killed, followed by exposure to fire, flood or natural disaster, being involved in a life-threatening accident and combat exposure. Molestation is more common in females than males. Mugging is more common in males than females. But in both instances women develop more PTSD. The only trauma where men develop more PTSD is rape.

PTSD is more common in younger than older individuals; more common in those with higher anxiety response to the initial event and in those who perceive external locus of control (as opposed to internal locus).

NICE encourages primary care diagnosis and screening, as PTSD is probably underdiagnosed.

Aetiology: Resilience to trauma is a dynamic factor; individuals who may not develop PTSD after one trauma may develop after another.

Factors associated with post-traumatic stress disorder (from Bisson, 2007) include p***re-traumatic factors such as p***revious psychiatric disorder, females gender, personality (external locus of control greater than internal locus of control), lower socioeconomic and educational status, ethnic minority status and personality disorders (cluster B); p***eritraumatic factors*** include the higher severity of trauma, perceived threat to life and peritraumatic dissociation. P***ost-traumatic factors include p***erceived lack of social support, subsequent life stress or physical illness (especially chronic pain). P***rotective factors*** include high IQ, higher social class, and getting an opportunity to grieve for the loss (e.g. viewing the dead body of friend/relative after trauma).

Hippocampus and amygdala show neuroimaging abnormalities. Hypocortisolaemia is reported in PTSD. Strong avoidance features may predict chronicity in PTSD.

Prevention of PTSD:

What are the effects of interventions to prevent post-traumatic stress disorder (Bisson, Clinical evidence 2004)**?**

Likely to be beneficial	Multiple-session CBT to prevent PTSD in people with acute stress disorder (reduced PTSD compared with supportive counselling)
Unknown effectiveness	Multiple-session CBT to prevent PTSD in all people exposed to a traumatic event Propranolol to prevent PTSD Single-session group debriefing to prevent PTSD Temazepam to prevent PTSD
Unlikely to be beneficial	Single-session individual debriefing to prevent PTSD Supportive counselling to prevent PTSD

(Prescriber 2005; 15(8):40-48, Ballanger et al., J Clin Psychiatry 2000; 61 (Suppl 5):60-66)

Treatment of PTSD:

NICE suggests the following:

Initial Management in Primary Care

- Watchful waiting where symptoms are mild and have been present for less than 4 weeks after the trauma
- The prescription of a non-benzodiazepine sleeping tablet after four consecutive nights' sleep disturbance is recommended

Further Specialist Management:

- Psychological treatment should be done in a regular and continuous (usually at least once a week) manner and should be delivered by the same person
- Non-trauma-focused interventions such as relaxation or non-directive therapy should not be used routinely since these do not address traumatic memories
- Interventions used when symptoms are present within 3 months of a trauma
    - trauma-focused cognitive behavioural therapy which includes a combination of exposure therapy, cognitive therapy and stress management
    - Should be offered to those with severe post-traumatic symptoms or with severe PTSD in the first month after the traumatic event and people who present with PTSD within 3 months of the event
    - Normally provided on an individual outpatient basis; around 8–12 sessions are usually offered (five sessions if the treatment starts within 1 month of the event)

    - The prescription of a nonbenzodiazepine medication after four consecutive nights of sleep disturbance is recommended. For short-term use - hypnotic medication; For longer-term use – an early introduction of a suitable antidepressant should be done to avoid dependence.
- Interventions used when symptoms are present for more than 3 months after a trauma
    - trauma-focused CBT or eye movement desensitisation and reprocessing (EMDR)
    - Up to 12 sessions can be offered. In case of treatment failure or limited improvement with a specific trauma-focused psychological treatment, consider an alternative form of trauma-focused psychological treatment; if no improvement considers pharmacological treatment.
    - Paroxetine, mirtazapine for general use; amitriptyline or phenelzine for specialist use.

Drug	Evidence & practice in the UK
Paroxetine	Good RCT evidence. NICE second line. Licensed for PTSD
Sertraline	RCT evidence; but NICE appraisal did not show significance. Licensed for females not males with PTSD in the UK.
Fluoxetine	1 RCT but not significant
Imipramine & Amitriptyline	Poor quality of evidence; but a statistically significant result for Amitriptyline; not so for imipramine.
Phenelzine	Poor quality of evidence; but statistically significant result
Mirtazapine	One small strongly positive RCT. NICE second line.
Venlafaxine	One large RCT no benefit
Olanzapine	Monotherapy RCT negative; augmentation of SSRIs positive
Risperidone	Tested only as adjunct – no effect

Considerable research has been conducted in particular approaches to the psychotherapy of PTSD. The evidence indicates that modalities tested in Randomised Controlled Trials are far from 100% applicable and effective. The RCT model itself is inadequate for evaluating treatments of conditions with complex presentations and multiple comorbidities. Psychotherapy studies often exclude patients with comorbidity and complexity (Corrigan & Hull, 2015)

In a meta-analysis by Bisson et al. (2007) no evidence of a difference in efficacy between TFCBT and EMDR was noted, but there was some evidence that TFCBT and EMDR were superior to stress management and other therapies, and that stress management was superior to other therapies. In a Cochrane review it was shown that psychological debriefing after trauma is either equivalent to, or worse than, control or educational interventions in preventing or reducing the severity of PTSD, depression, anxiety and general psychological morbidity after trauma. There is some suggestion that it may increase the risk of PTSD and depression.

Trauma-focused CBT: May includes e*xposure therapy* wherein repeated confrontation of traumatic memories and repeated exposure to avoided situations take place together with relaxation and anxiety reduction. In trauma-focused *cognitive component* modification of misinterpretations that lead to overestimation of current threat and modification of other beliefs related to the traumatic experience and the individual's behaviour during the trauma (for example, guilt and shame) are attempted via cognitive restructuring process. Trauma-focused psychological treatment should usually be given for eight to 12 sessions

Eye movement desensitisation and reprocessing: This was serendipitously discovered by a psychologist (Shapiro) when she first applied it to herself. It is based on the theory that bilateral stimulation, in the form of eye movements, allows the processing of traumatic memories. While the patient focuses on specific images, negative sensations and associated cognitions, bilateral stimulation is applied to desensitize the individual to these memories and more positive sensations and cognitions are introduced.

Outcome: More than a third of people reported having the disorder six years after they first developed it. 50% chance of remission at two years.

Acute stress disorder is a diagnostically different entity from PTSD. To diagnose acute stress reaction, there must be an immediate and clear temporal connection between the impact of an exceptional stressor and the onset of symptoms. The symptoms usually appear within minutes of the impact of the stressful stimulus or event and disappear within 2-3 days (often within hours). Partial or complete amnesia for the episode may be present. In addition, the symptoms:
(a) Show a **mixed** and usually **changing picture**; in addition to the initial state of "daze", depression, anxiety, anger, despair, overactivity, and withdrawal may all be seen, but no one type of symptom predominates for long;
(b) **Resolve rapidly** (within a few hours at the most) in those cases where removal from the stressful environment is possible; in cases where the stress continues or cannot by its nature be reversed, the symptoms usually begin to diminish after 24-48 hours and are usually minimal after about 3 days.
This diagnosis should not be used to cover sudden exacerbations of symptoms in individuals already showing symptoms that fulfill the criteria for any other psychiatric disorder, except for those in F60 (personality disorders). However, a history of previous psychiatric disorder does not invalidate the use of this diagnosis.

4. Generalised Anxiety Disorder:

The feature of avoidance seen in other anxiety disorders is not needed for a diagnosis of GAD; it is often punctuated by depressive episodes and co-existent with a number of comorbid diseases.

Epidemiology: the lifetime prevalence of DSM-IV GAD is estimated to be about 5%; point prevalence is about 2% to 3% (Weisberg, 2009)

Aetiology: Twin studies suggest MZ concordance 41% vs. DZ 4%, though the rates vary in different samples (e.g. Andrews et al. study - 21.5% concordance in MZ twins, compared with a 13.5% in DZ). Risk factors include exposure to civilian trauma, bullying or peer victimisation. a higher number of life events, being a first-degree relative of a GAD patient and female gender.

Hamilton anxiety scale is a 14-item instrument that emphasizes somatic symptoms. Treatment response is generally defined as a 50% reduction in baseline score. Clinical recovery is often defined as a score of less than 7 on the Hamilton anxiety scale.

Treatment:

Acute treatment	SSRIs: escitalopram, paroxetine, sertraline. TCAs: imipramine. Benzos: alprazolam and diazepam (not suited for long-term therapy). Venlafaxine, duloxetine & Buspirone CBT
Long-term efficacy/relapse prevention	Paroxetine, escitalopram, CBT, venlafaxine, pregabalin.
Adjuncts for non-response	Antipsychotics olanzapine/risperidone low doses.

Adapted from Tyrer, P & Baldwin, D (2006). Generalised anxiety disorder. 368, 9553: 2156-2166

NICE recommends that an SSRI should be used as a first line. SNRIs and pregabalin are alternative choices. Among psychological treatments, both CBT and anxiety management therapy are useful though CBT may be better in efficacy. CBT includes education, relaxation training, and exposure and cognitive restructuring.

Guidelines including NICE suggest that there is insufficient evidence to recommend combined psychological and pharmacological treatment initially, although this can be considered if initial treatment fails.

Outcome: At 12-year follow-up of adults at an anxiety clinic, 42% of patients had recovered from generalised anxiety disorder, but the disorder was a marker for poor outcome in those patients who had another anxiety disorder.

Herbal Treatment of Anxiety: The kava shrub (*Piper methysticum*) has been studied in anxiety disorders and appears to be effective in some trials. Gingko and valerian are not effective; common lavender, saffron, German chamomile and lemon balm appear promising ***(Ernst 13 (4): 312. APT (2007)***.

- The effects of kava are due to the kavapyrones, which in animal models act as muscle relaxants and anticonvulsants, protect against strychnine poisoning, and reduce limbic system excitability.
- They may cause inhibition of voltage-dependent sodium channels, increase GABAA receptor densities, block norepinephrine reuptake, and suppress the release of glutamate.
- Several double-blind, placebo-controlled trials have shown kava to be more effective than placebo in reducing HAM-A scale scores, and this effect is detectable even within 1 week of therapy. Werneke *et al. in* found that kava (*Piper methysticum*) was the most researched remedy for anxiety and that there was good evidence for its anxiolytic effect. A Cochrane review reported of 11 RCTs showed that kava is the only herbal remedy that has been proven to be effective in reducing anxiety. But kava cannot be recommended for clinical use because of an association with hepatotoxicity, which has led to its withdrawal from the UK market.
- Kava is associated with allergic, sometimes fatal hepatotoxicity and so not recommended for general use. (Beaubrun & Gray, 2000)
- Cochrane reviews of valerian and passiflora showed no efficacy in anxiety compared to diazepam.
- Kava can interact with levodopa and alprazolam, causing extrapyramidal symptoms or lethargy. Valerian can interact with loperamide and fluoxetine, causing delirium. Evening primrose oil can interact with phenothiazides, causing epileptic seizures.

5. Social Phobia:

Point prevalence is 2.8%. Two types are recognised (Schneier, 2003): A g**eneralised** subtype where fear occurs in most social situations. This leads to higher impairment and patients are more likely to seek treatment, than the second subtype. A s**ituational** or non generalised subtype presents as fear of public speaking or performance anxiety.

Treatments: The best-established treatments are **CBT & SSRIs**. Benzodiazepines should not be used.

- Paroxetine, sertraline, fluoxetine and fluvoxamine, escitalopram, venlafaxine have been evaluated in RCTs with a favourable outcome. Normal antidepressant doses can be used but may need a longer 12-week trial to establish outcome. Drug treatment should continue for at least 6-12 months after the response.
- **Phenelzine** is the **second line** treatment. Moclobemide is also effective.
- Short-term use of benzodiazepines including clonazepam and novel anticonvulsant gabapentin has been evaluated with some evidence.
- **SSRI + clonazepam** combination, **gabapentin or pregabalin** can be tried as **third-line** agents.
- β Blockers are not significantly different from placebo in generalised subtype, but they can be used on/off in performance anxiety subtype.
- **Self-help principles** should be encouraged.

6. Panic Disorder:

Panic can exist in different forms. Major classificatory systems recognise panic disorder, agoraphobia and comorbid panic disorder with agoraphobia. DSM puts panic disorder as primary dysfunction while ICD focuses on agoraphobia. To diagnose panic disorder, there must be frequent panic attacks within a specified time interval.

Epidemiology: It is increasingly being realised that panic attacks can occur without fully satisfying panic disorder criteria. Patients with such panic attacks have a significant impairment, help seeking and medication requirement. Point prevalence of panic disorder is 0.9%. The NCS-R collected data on four composite groups: Isolated panic attacks (PA only), Panic attacks with AG (PA-AG), Panic disorder (PD only), PD with AG (PD-AG). Lifetime prevalence of PA only was 28% **while 4.7% had lifetime PD. O**nly 1.1% had PD-AG while 0.8% had lifetime PA-AG. Cued panic attacks are common in PD and AG compared to isolated PA where non-cued out of blue panic is seen. The mean age of onset of any panic attack – irrespective of diagnosis – is around 22 years. All-cause mortality is increased by 1.9 times (SMR) in those with panic disorder.

The ICD-10 classifies panic disorder as recurrent, unpredictable panic attacks, with sudden onset of palpitations, chest pain, choking sensations, dizziness, and feelings of unreality, often with associated fear of dying, losing control, or going mad, but without the requirement for the symptoms to have persisted for 1 month or longer.

Aetiology: An estimated heritability of 30%-40% is noted. According to cognitive theory patients with panic disorder have a heightened sensitivity to internal bodily sensations. A fear network involving the amygdala, orbitofrontal cortex and hypothalamus has been implicated in some neuroimaging studies.

Treatment: NICE guideline summary

- **CBT psychotherapy** - seven to 14 hours of cognitive behaviour therapy—usually weekly sessions of one to two hours, completed within four months is recommended.
- **Self-help by bibliotherapy** and self-help groups and support groups are useful in primary care.
- **SSRIs** are useful **first-line drug treatments**.
- If no improvement is seen after a 12-week course of SSRIs, imipramine or clomipramine should be considered.
- Benzodiazepines are associated with a worse outcome in the long term and should not be prescribed.

BAP recommendations for panic disorder:

For **acute management** according to patient choice:

- ***pharmacological***: all SSRIs, some TCAs (clomipramine, imipramine), some benzodiazepines (alprazolam, clonazepam, diazepam, lorazepam), venlafaxine, reboxetine have evidence base or ***psychological***: cognitive-behaviour therapy.
- Both pharmacological and psychological approaches have broadly similar efficacy in acute treatment.

- Consider an SSRI for first-line pharmacological treatment.
- Consider increasing the dose if there is an insufficient response, but there is only limited evidence for a dose-response relationship with SSRIs.
- Treatment periods of up to 12 weeks are needed to assess efficacy.

Longer-term treatment

- Consider cognitive therapy with exposure as this may reduce relapse rates better than drug treatment.
- Continue drug treatment for a further six months in patients who are responding at 12 weeks: first line drug choice is an SSRI; imipramine is a second-line choice
- Routinely combining drug and psychological approaches is not recommended.

When initial treatments fail

- Consider switching at 12 weeks
- Consider combining evidence-based treatments only when there are no contraindications Consider adding paroxetine or buspirone to psychological treatments after partial response
- Consider adding paroxetine, whilst continuing with CBT, after initial non-response.

7. Other related disorders

A. Hypochondriasis

Classed in ICD-10 as a preoccupation with the fear of having a serious disease based on the misrepresentation of bodily symptoms. DSM-V has removed hypochondriasis as a disorder as the name was perceived as pejorative and most patients with a former diagnosis of hypochondriasis would now be diagnosed as having Somatic Symptom Disorder or Illness Anxiety Disorder (see below).

Prevalence and Risk Factors

Most information comes from studies conducted in primary care. Prevalence has been reported between 0.8 and 4.5%. There is no conclusive data about specific risk factors, but there are associations with anxiety, depressive and other somatoform disorders.

Treatment

CBT has been shown to be efficacious, and group cognitive-behavioural therapy may also be useful. SSRIs may also be efficacious.

B. Body Dysmorphic Disorder:

A patient with body dysmorphic disorder (BDD) (dysmorphophobia) is **convinced that part of their body has a defect or is flawed in some way**. To the observer, the appearance is normal or of a minor abnormality. The patient engages in repetitive behaviours or mental acts in response to preoccupations with perceived defects or flaws. BDD has both **psychotic and non-psychotic variants**, classified in DSM-V as BDD with delusional or without delusional component, suggesting they can be considered as part of the same disorder, characterized by a spectrum of insight.

Aetiology: Factors that may **predispose** persons to BDD include:

- low self-esteem
- critical parents and significant others
- early childhood trauma
- unconscious displacement of emotional conflict

Epidemiology: Patients with BDD also had an earlier onset of major depression and higher lifetime rates of major depression (26%), social phobia (16%), obsessive-compulsive disorder (6%), and psychotic disorder diagnoses as well as higher rates of substance use disorders in first-degree relatives. Studies have reported rates of BDD of 7% and 15% of patients seeking cosmetic surgery and a rate of 12% in patients seeking dermatologic treatment.

Treatment: High-dose SSRIs used for a longer duration than the usual antidepressant trial period can be effective. Fluoxetine has RCT evidence in this regard. Antipsychotics have also been studied with variable success. Pimozide has anecdotal support though the risk of QT prolongation precludes wider use. CBT has been shown to be effective in a number of case series; combination with fluoxetine may treat resistant cases.

Outcome: The prognosis of psychotic variant is generally poor with waxing and waning course. But there is often a relatively preserved psychosocial functioning in many patients, in line with the outcomes seen in persistent delusional disorders.

C. Medically Unexplained Symptoms and Somatisation Disorder:

Somatisation Disorder: Community surveys in the United States and Western Europe have found prevalence rates of less than 1 per cent. Primary care surveys using structured interviews have found only a slightly higher prevalence—typically **1 to 2%**. An analysis of follow-up data from the World Health Organization multicentre primary care survey, however, indicates that recall during structured interviews may significantly underestimate the lifetime prevalence of somatization disorder and unexplained somatic symptoms. The prevalence of somatization disorder and of less severe somatization syndromes is typically twice as high in women as men. The survey also noted that approximately 20 percent of primary care patients suffered from persistent pain (one or more pain symptoms present for most of the last 6 months). Pain syndromes are approximately twice as prevalent in women as in men.

A high proportion of patients with MUS have undiagnosed and, therefore, untreated mental disorder. Rohricht and Elanjithara (2009) reported that 42% of those with MUS had a primary diagnosis of somatoform disorder, 36% had a depressive disorder and depressive symptoms medicated by the effect of somatic symptoms.

Presentation: The reliability of medically unexplained symptoms is limited; they are a feature of Somatic Symptom Disorders (DSM-V); Somatic Symptom Disorders can also accompany diagnosed medical disorders. The emphasis on a diagnosis of Somatic Symptom Disorder is on the maladaptive thoughts, feelings, and behaviours associated with the somatic symptoms.

Treatment: A review of studies carried out in primary, secondary, and tertiary care settings by Sumathipala et al 2007 showed three types of interventions to be supported by evidence for medically unexplained symptoms, namely: **antidepressant medication**, **cognitive behavioural therapy** (CBT), and other **nonspecific interventions**.

- There is more level I evidence for CBT compared to other approaches.
- CBT is efficacious for the broader category of medically unexplained symptoms by reducing physical symptoms, psychological distress, and disability.
- No studies have compared pharmacological and psychological treatments.
- Most trials assessed only short-term outcomes.

New and locally derived **collaborative care models** of active engagement in primary care settings are recommended. Patients with somatoform disorder may benefit from Body-Oriented Psychological Therapy, mentalization-based CBT, and brief psychodynamic interpersonal therapy.

D. Dissociative Disorders:

Dissociative disorders refer to a set of conditions that involve disruptions or breakdowns of memory, identity, or perception. ICD-10 classifies Conversion Disorder as a dissociative disorder. DSM-V lists: Dissociative Identity Disorder (the distinct alternation of two or more distinct personality states with impaired recall among personality states; and Dissociative Amnesia (the temporary loss of recall memory, specifically episodic memory, due to a traumatic or stressful event. Classed under this is Dissociative Fugue (reversible amnesia for personal identity, usually involving unplanned travel or wandering, sometimes accompanied by the establishment of a new identity). Included in the Dissociative Disorders is Depersonalisation/derealisation disorder.

Prevalence: Some surveys estimate that 10% of the adult population has dissociative disorder (according to DSM-III definition). It appears to be as common as anxiety, mood and substance misuse disorders. It may be that more women present with the severest symptoms, but several studies have found no gender differences in the prevalence of Dissociative Disorders.

Aetiology: Patients with Dissociative Disorders report high frequencies of childhood psychological trauma, in particular childhood sexual abuse (57-90%), emotional abuse (57%), physical abuse (62-82%) and neglect (62%).

Treatment: The aim is to integrate feelings, perceptions, thoughts and memories. The international Society for the Study of Trauma and Dissociation recommend individual psychotherapy (especially **structured therapy** such as Acceptance and Commitment Therapy and DBT) and skills training, although studies in this field are in their infancy.

8. Eating Disorders:

3 major types are recognised by ICD-10: Anorexia nervosa, bulimia nervosa and EDNOS – atypical ED or ED not otherwise specified. Binge eating disorder is the increasingly recognised fourth type, but falling under EDNOS currently. But the stability of individual ED diagnoses is poor; patients migrate from one to other very often. Notably at least 1/4th to 1/3rd of those with bulimia have a past history of anorexia though migration from bulimia to anorexia is somewhat rare.

In the diagnostic criteria for Anorexia Nervosa (AN) the value of the amenorrhoea criterion is questionable as most female patients who have BMI around 17.5 and body image disturbance are amenorrhoeic; those who have intact periods have same clinical features and outcome as those who satisfy all 3 criteria. In DSM-5, the requirement for amenorrhoea has been eliminated.

In bulimia binge eating episodes are characteristically seen (may also occur in anorexia). DSM-V specifies a once-weekly frequency for binge eating and inappropriate compensatory behavior as a diagnostic criterion. The amount of food intake during binges varies from 1000 kCals to 2000 kCals. Most patients with bulimia have an immense feeling of loss of control and so seek and engage in treatment better than anorexia patients.

Epidemiology of bulimia and anorexia:

Anorexia	Bulimia
Onset mostly in adolescents	Mostly young adults; little later onset
Excess in higher social class	Even class distribution
0.5-1% prevalence in teenage girls	1-2% prevalence in 16-35 age group
19/100 000 females per year	29/100 000 females per year

(adapted from Fairburn & Harrison, 2003)

65% patients with anorexia have depression; 34% have social phobia while 26% have OCD.

Aetiology: Shared familial liability is noted among the three EDs. EDs are also associated with certain personality traits. Substance use is increased in families of bulimic probands; obsessional and perfectionist traits are increased in families of anorexic probands.

	Monozygotes	Dizygotes
Anorexia nervosa	**55%**	5%
Bulimia nervosa	35%	30%

A significant heritability exists for anorexia nervosa but not for bulimia nervosa.

Risk factors: (adapted from Fairburn & Harrison, 2003)

- Female sex, adolescence and early adulthood
- Western cultural adaptation
- Family history of ED, depression, substance misuse, especially alcoholism (for bulimia nervosa), obesity (for bulimia nervosa)
- Adverse parenting (especially low contact, high expectations, parental discord), childhood sexual abuse, critical comments about eating, shape, or weight from family and others
- Occupational and recreational pressure to be slim
- Low self-esteem, perfectionism (anorexia nervosa more than bulimia nervosa)
- Past history of being obese (bulimia nervosa)
- Early menarche (bulimia nervosa)

Binge eating disorder (BED) is characterized by **r**ecurrent episodes of binge eating in the absence of extreme weight-control behaviour. There is often a background of a general tendency to overeat. An association with obesity is seen; 5–10% of those seeking treatment for obesity have BED. Patients typically present in their 40s, with more males compared to other EDs, but only 25% of all binge-eating population are males. A high degree of spontaneous remission is seen; stress associated overeating is a common phenomenon in those with BED. Self-help, behavioural weight loss programmes and CBT/IPT can help.

Physical symptoms of EDs:

- Increased sensitivity to cold
- Gastrointestinal symptoms—e.g., constipation, fullness after eating, bloatedness
- Dizziness and syncope
- Amenorrhoea (in females not taking an oral contraceptive), low sexual appetite, infertility
- Poor sleep with early morning wakening

Physical signs of EDs:

- Emaciation; stunted growth and failure of breast development (if prepubertal onset)
- Dry skin; fine downy hair (lanugo) on the back, forearms, and side of the face; in patients with hypercarotenaemia, orange discolouration of the skin
- Russel's sign – calluses in knuckles due to repeated vomit induction
- Swelling of parotid and submandibular glands (especially in bulimic patients)
- Erosion of inner surface of front teeth (perimylolysis) in those who vomit frequently
- Cold hands and feet; hypothermia
- Bradycardia (heart rate around 40 is common); orthostatic hypotension; cardiac arrhythmias (especially in underweight patients and those with electrolyte abnormalities). Systolic BP may drop up to 70 in the extremely starved.
- Dependent oedema (complicating assessment of bodyweight)
- Weak proximal muscles (elicited as difficulty rising from a squatting position)

Abnormalities on physical investigation:

- Endocrine
    - Low concentrations of luteinizing hormone, follicle stimulating hormone, and oestradiol
    - Low T3, T4 in low normal range, normal concentrations of thyroid stimulating hormone (low T3 syndrome)
    - Mild increase in plasma cortisol
    - Raised growth hormone concentration
    - Severe hypoglycaemia (rare)
    - Low leptin (but possibly higher than would be expected for bodyweight)
- Cardiovascular
    - ECG abnormalities (especially in those with electrolyte disturbance): conduction defects, especially prolongation of the Q-T interval, of major concern
    - Ipecac (emetic substance) contains emetine an alkaloid that can cause myopathy and fatal cardiomyopathy which may be reversible in early stages.
- Gastrointestinal
    - Delayed gastric emptying
    - Decreased colonic motility (secondary to chronic laxative misuse)
    - Acute gastric dilatation (rare, secondary to binge eating or excessive re-feeding)
- Haematological
    - Moderate normocytic normochromic anaemia
    - Mild leucopenia with relative lymphocytosis
    - Thrombocytopenia
- Other metabolic abnormalities
    - Hypercholesterolaemia
    - Raised serum carotene
    - Hypophosphataemia (exaggerated during refeeding)
    - Dehydration
    - Electrolyte disturbance (varied in form; present in those who vomit frequently or misuse large quantities of laxatives or diuretics): vomiting results in metabolic alkalosis and hypokalaemia; laxative misuse results in metabolic acidosis, hyponatraemia, hypokalaemia
- Other abnormalities
    - Osteopenia and osteoporosis (with heightened fracture risk)
    - Enlarged cerebral ventricles and external cerebrospinal fluid spaces (pseudoatrophy)

Effects on pregnancy:

- Decreased fertility – but regained near normally on complete recovery.
- May have more abortions
- Higher rates of hyperemesis gravidarum, anaemia, impaired weight gain
- Compromised intrauterine foetal growth
- Premature delivery is more likely
- Rates of caesarean delivery are high
- Post-natal complications and post-partum depression are higher
- Associated with low birth weight, microcephaly, low APGAR scores.
- In actively anorexic mother, the neonate may have hypoglycaemia

Evidence-based management of eating disorders:

Evidence	**Anorexia nervosa**	**Bulimia nervosa**
Antidepressants (acute treatment)	Modest	Considerable
Antidepressants (relapse prevention)	Modest	Modest
Cognitive analytic therapy (CAT)	Modest	None
Cognitive behaviour therapy	Modest	Strong
"Dialectical behaviour therapy"-based treatment	None	Modest
Family-based therapy for adolescents	Moderate	None
Interpersonal psychotherapy (IPT)	None	Moderate

(adapted from Fairburn & Harrison, 2003)

Managing bulimia:

- Most effective treatment: Cognitive behaviour therapy.
- Typically involves about 20 individual treatment sessions over 5 months;
- 33-50% make a complete and lasting recovery.
- Antidepressant drugs have an antibulimic effect.
- Produce a rapid decline in the frequency of binge eating and purging, and an improvement in mood.
- Not as effective as CBT but.
- The effect is often not sustained.

Managing anorexia:

Major therapeutic goals:

1. engagement
2. weight restoration
3. psychological therapy – cognitive restructuring
4. if needed use of compulsion

Outpatient therapy for anorexia has best chance if

a. Illness is present for less than 6 months
b. No bingeing or vomiting
c. Have parents who cooperate and are willing to participate in family therapy

Summary of NICE guidelines
Anorexia: ▪ Drugs should not be used as sole or primary treatment for anorexia nervosa ▪ For anorexia nervosa, consider cognitive analytic or cognitive behavioural therapies, interpersonal psychotherapy, focal dynamic therapy, or family interventions focused on eating disorders. ▪ Family interventions that directly address the eating disorder are especially useful for children and adolescents with anorexia nervosa ▪ Dietary counselling should not be provided as sole treatment for anorexia nervosa
Bulimia: ▪ Evidence-based self-help programme is first line for bulimia; as an additional option or alternative choice antidepressants can be used in bulimia. ▪ Antidepressant drugs can reduce frequency of binge eating and purging, but long-term effects are unknown; ▪ Any beneficial effects of antidepressants will be rapidly apparent ▪ SSRIs (specifically fluoxetine) are drugs of first choice for bulimia nervosa Effective dose of fluoxetine is higher than for depression (60 mg daily) ▪ Specifically adapted cognitive behavioural therapy should be offered to adults with bulimia nervosa, 16–20 sessions over 4–5 months ▪ Interpersonal psychotherapy should be considered as alternative to cognitive behavioural therapy, but patients should be informed it takes 8–12 months to achieve similar results
Binge eating disorder has similar lines of management guidelines to bulimia.

http://apt.rcpsych.org/content/18/1/34

9. Personality Disorders:

Epidemiology: Personality disorders, as a group, are among the most frequent disorders treated by psychiatrists (Zimmerman et al., 2005). Epidemiological studies suggest that between 5% and 13% of the population has at least one personality disorder.

In general, the prevalence of personality disorders among **psychiatric outpatients** and **inpatients** is high, with many studies reporting a prevalence of greater than 50% of samples. Borderline PD is generally the most prevalent in psychiatric settings. **Personality disorders are particularly prevalent among inpatients with drug, alcohol, and eating disorders**. In these populations, prevalence figures for personality disorder have been reported to be **in excess of 70%.**

Dissocial personality is the most prevalent category of personality disorder in **prison** settings. A survey of a randomly selected sample of one in six prisoners in England and Wales, found that the prevalence of any personality disorder was **78% for male remand, 64% for male sentenced and 50% for female prisoners** . Antisocial personality disorder had the highest prevalence of any category of personality disorder, with 63% of male remand prisoners, 49% of sentenced prisoners and 31% of female prisoners. In Great Britain, the **prevalence of antisocial PD in general population is estimated at 0.6%** (See Coid et al – Tables below), with the **rate in men (1%) five times that in women (0.2%).** Surveys conducted in other countries report prevalence rates for ASPD ranging from 0.2% to 4.1%. Higher prevalence rates for personality disorders appear to be found in urban populations, and this may account for some of the range in reported prevalence.

Personality disorder	Median prevalence rate per 1000 in community **(from Oxford Textbook of Psychiatry, 2000)**	Weighted prevalence per 1000 in community sample **(from Coid 2006 BJPsych – Great Britain data)**	Prevalence in a psychiatric outpatient sample per 1000 **(Zimmerman et al. 2005)**
Paranoid	6	7	42
Schizoid	4	8	14
Schizotypal	6	0.6	6
Antisocial	19	6	36
Borderline	16	7	93
Histrionic	20	12	10
Narcissistic	2	8	23
Anankastic	17	19	87
Avoidant	7	8	147
Dependent	7	1	14
Passive aggressive	17	-	No data
		Cluster A = 1.6% Cluster B = 1.2% Cluster C = 2.6% Any PD = 4.4%	Cluster A = 6% Cluster B = 13% Cluster C = 22% Any PD = 46%

Borderline personality disorder is seen in nearly 2% of general population, 10% outpatients, 15-20% inpatients and 30-60% of all personality disorders in some clinical samples. The estimated female: male ratio is 3:1. It is 5 times more common in first-degree relatives of borderline patients.

Diagnostic stability:

- For a long time it was thought that once diagnosed with personality disorder the diagnosis remains forever, and no change takes place. This pessimism is now shifting largely due to long-term follow-up studies in borderline personality.
- **Longitudinal course of Borderline Personality**: 2 important studies to date are the McLean study and CLPDS study. In McLean Study of Adult Development, the **prevalence of five core borderline symptoms was found to decline with particular rapidity: quasi-psychotic thought, self-mutilation, help-seeking suicide efforts, treatment regressions, and countertransference problems**. In contrast, feelings of depression, anger, and loneliness/emptiness were the most stable symptoms. The 10 year follow-up of McLean sample further clarified this. **Chronic dysphoria, intense anger, nondelusional paranoia, general impulsivity** (disordered eating, spending sprees, or reckless driving**) remained relatively common even after 10 years of prospective follow-up** while other features largely abated. In the Collaborative Longitudinal Personality Disorders Study, abandonment fears and physically self-destructive acts were found to be the least stable (rapidly remitting).
- Results of epidemiological studies suggest a J-shaped relation between personality and age, with an initial decrease followed by an increase in some personality disorders in older people.
- Seivewright & Tyrer reported a 12-year follow-up where 178 out of 202 patients were reassessed for the personality status. The personality traits of patients with the cluster B [flamboyant group - antisocial, histrionic] became significantly less pronounced over 12 years, but those in the cluster A (odd, eccentric group - schizoid, schizotypal, paranoid), and the cluster C (anxious, fearful group - obsessional, avoidant) became more pronounced.
- Evidence from the McLean Study of Adult Development suggests that 40% of patients with borderline personality disorder remit after 2 years, with 88% no longer meeting criteria after 10 years (Silk, KR Am J Psychiatry 165:413-415, April 2008).

<u>Diagnostic features in a nutshell:</u>

Cluster A (odd, eccentric)
Paranoid personality disorder: ▪ Suspicious of other people and their motives. ▪ hold longstanding grudges against people, ▪ believe others are not trustworthy, ▪ emotionally detached ▪ Feel other people are deceiving, threatening, or making plans against them. Schizoid personality disorder: ▪ Have difficulties in expressing emotions, particularly around warmth or tenderness. ▪ Prefer loneliness

- aloof or remote,
- have difficulty in developing or maintaining social relationships
- remain unaware of social trends
- unresponsive to praise or criticism

Schizotypal personality disorder (in ICD10 – classified as a type of schizophrenia)

- appear odd or eccentric;
- may have illusions, magical thinking
- obsessions without resistance
- may be members of quasi-cultural groups
- Thought disorders and paranoia.
- may believe in ESP, clairvoyance, etc
- may have transient psychotic features

Cluster B (dramatic, erratic)

Antisocial personality disorder:

- Lack of regard for the rights and feelings of other people.
- Lack of remorse for actions that may hurt others.
- ignore social norms about acceptable behaviour,
- May disregard rules and break the law.
- Make relations easily but break them equally easily
- A small proportion may be psychopathic

Borderline personality disorder:

- poor self-image,
- unstable personal relationships,
- Impulsive behaviour in areas such as personal safety and substance misuse.
- may self-harm, feel suicidal and act on these feelings,
- experience the instability of mood,
- Have episodes of micro-psychosis.
- feelings of chronic emptiness
- fears of abandonment – rejection sensitivity hence form intense but short lasting relations

Histrionic personality disorder:

- Extreme or over-dramatic behaviour.
- may form relationships quickly, but be demanding
- Attention-seeking.
- may appear to others as being self-centred,
- having shallow emotions,
- Being inappropriately sexually provocative.

Narcissistic personality disorder:

- An exaggerated sense of own importance.
- frequently self-centred
- Intolerant of other people.
- grandiose plans and ideas
- Cravings for attention and admiration.

Cluster C (anxious, inhibited)
Avoidant personality disorder ▪ fears being judged negatively by other people ▪ Feelings of discomfort in a group or social settings. ▪ may come across as being socially withdrawn ▪ Have low self-esteem. ▪ May crave affection but fears of rejection overwhelming. Dependent personality disorder ▪ assumes a position of passivity, ▪ Allowing others to assume responsibility for most areas of their daily life. ▪ lack self-confidence, ▪ feel unable to function independently of another person, ▪ Feels own needs are of secondary importance. Obsessive-compulsive personality disorder (1 -2% prevalence) ▪ Difficulties in expressing warm or tender emotions to others. ▪ frequently perfectionists ▪ often lack clarity in seeing other perspectives or ways of doing things, ▪ Rigid attention to detail may prevent them from completing tasks. ▪ Some may be hoarders, scrupulous with money ▪ May not be able to delegate tasks; workaholics.

Though presence of transient stress-induced paranoid ideations in borderline PD (DSM) qualifies for 'quasi-psychotic' label, the term 'quasi psychotic episode' is in fact used only for schizotypal disorder in the contemporary classificatory systems. The ICD10 schizotypal disorder is characterised by 'occasional transient quasipsychotic episodes with intense illusions, hallucinations and delusion-like ideas usually occurring without external provocation'.

Evidence for pharmacological management: (Adapted from Tyrer & Bateman, 2004)

- ***Amitriptyline*** - no effect on borderline disorder even if depressed; haloperidol compares better.
- ***Flupentixol*** – some efficacy on self-harm behaviour.
- ***Antipsychotics*** - low dosage these drugs might be effective in the treatment of both schizotypal and borderline personality disorders
- ***SSRIs*** – reduce aggressive, impulsive and angry behaviour in those with borderline and aggressive personality disorders.
- ***Anticonvulsants*** and ***lithium*** – some effect against affective dysregulation in borderline disorder and aggressive outbursts in cluster B.

Note that according to NICE Guidelines, pharmacological treatment is not recommended for the features of Personality Disorder, but it is for co-morbid illness (such as depression or OCD).

Type of symptom	Preferred drug treatment
Cognitive/perceptual Suspiciousness, paranoid ideation, ideas of reference, magical thinking, stress-induced hallucinations	**Antipsychotics**
Affective dysregulation Lability of mood, 'rejection sensitivity', mood crashes, temper outbursts, chronic emptiness, and dysphoria.	**SSRIs**
Impulsive-behavioural dyscontrol Sensation-seeking, risky or reckless behaviour, low frustration tolerance, impulsive aggression, impulsive binges, recurrent suicidal behaviour.	**Mood stabilisers, SSRIs**

Psychological therapies are considered with psychotherapy section.

Differentiating bipolar and borderline disorder:

Bipolar Disorder	Borderline Personality Disorder
Onset in teens or early 20s	No defined onset
Observable, spontaneous mood changes, last for days to weeks	Often not observable; changes are precipitated by internal or external events, last for hours
Dysphoric, anxious and elated mood shifts	Dysphoric, anxious and angry but elated mood is rare
Episodic impulsivity and risk-taking	Chronic impulsivity and risk-taking
Episodic suicide attempts related to depressive episodes	Recurrent suicidal gestures
Self-mutilation rare	Self-mutilation common
Endorse 'depressed mood' as descriptor	Endorse 'emptiness' as descriptor
Family history of bipolar I or II or recurrent depression	Family history negative for bipolar I, II and recurrent depression

Adapted from Yatham L, et al. *Bipolar Disord* 2005: 7 (Suppl. 3): 5–69.

Richardson & Tracy (2015) highlight six core illness-differentiating themes influencing patient perspectives on these two conditions:

- **Public information**: for bipolar there is greater public awareness, positive celebrity exposure, more public information and support groups
- **Delivery of diagnosis**: this is perceived to be taken more seriously and be given more time by staff than BPD
- **Illness causes**: perceived to be more genetic, and to do with brain 'wiring' or 'chemical' problem in bipolar than in BPD

- ✶ **Illness management**: more medication-orientated in bipolar disorder, better-established protocols in comparison with BPD
- ✶ **Stigma and blame**: greater stigma for BPD, with accompanying feeling of staff hopelessness and self-blame
- ✶ **Relationships with others**: a diagnosis of bipolar is accompanied with greater support from family and friends; its infrequent nature makes it less troublesome and easier to conceal, whereas BPD is associated with insidious destruction and sabotage of relationships, felt to be ever-present and cannot be concealed from relationships

10. Psychosexual disorders

Sexual dysfunctions are coded in F50 group in ICD10. They can be classified as (according to the terms used in DSM)

- Sexual desire disorders (sexual aversion, hypoactive sexual desire)
- Sexual arousal disorder (female sexual arousal disorder, male erectile disorder)
- Orgasmic disorders (female and male orgasmic disorder, premature ejaculation)
- Sexual pain disorder (Dyspareunia, vaginismus)
- Others (including those due to general medical or substance use disorders)

Hypoactive sexual desire is the most common female sexual dysfunction. It is the most difficult to treat too. Testosterone has been tried as a treatment in both males and females but with modest effects, largely due to side effects.

DSM-5 specified paraphilic disorders (from Yakeley & Wood, 2014)

- Voyeurism (spying on others' sexual/ private activities)
- Exhibitionism (deliberately exposing one's genitals)
- Frotteurism (deliberately rubbing against a non-consenting individual)
- Sexual masochism (choosing to undergo humiliation, bondage or suffering to achieve sexual arousal)
- Sexual sadism (choosing to inflict humiliation, bondage or suffering on one's partner to achieve sexual arousal)
- Paedophilia (sexual focus on children)
- Fetishism (use of objects or focussing on non-genital body parts to achieve arousal)
- Transvestic disorder (engaging in sexually arousing cross-dressing)

'Other specified paraphilic disorder': includes zoophilia (animals), scatalogia (obscene phone calls), necrophilia (corpses), coprophilia (faeces), klismaphilia (enemas), urophilia (urine)

Notes prepared using excerpts from

- Baldwin et al (2005) Evidence based guidelines for management of anxiety disorders. J of Psychopharmacology. 19; 567-596.
- Beaubrun, G & Gray, GE (2000) A review of herbal medicines for psychiatric disorders. Psychiatric Services 51:1130–1134, 2000
- Bisson JI. Post-traumatic stress disorder. BMJ 2007;334:789-93
- Bisson, J et al. Psychological treatments for chronic post-traumatic stress disorder: Systematic review and meta-analysis The British Journal of Psychiatry 2007; 190: 97-104
- Castle, DJ & Phillips, KA. Obsessive compulsive spectrum of disorders: A defensible construct? Aust N Z J Psychiatry. 2006; 40(2): 114–120.
- Diagnostic and Statistical Manual of Mental Disorders (Fifth text revision ed.). American Psychiatric Association, Washington DC 2014
- Fairburn & Harrison (2003) Eating disorders. The Lancet. 361, 9355; 407-416.
- Gale, C & Davidson, O. Generalised anxiety disorder. BMJ 2007;334:579-581
- **Gunderson, JG. "Borderline Personality Disorder: Ontogeny of a Diagnosis," *Am J Psychiatry* 166, no. 5 (May 1, 2009): 530-539.**
- Gunstad J, Phillips KA. Axis I comorbidity in body dysmorphic disorder. Compr Psychiatry. 2003;44:270-6.
- Heyman, I et al. Obsessive compulsive disorder. BMJ 2006;333(7565):424
- http://www.medscape.com/viewarticle/431268_4
- Human Reproduction Update 2006 12(3):193-207
- Jones, W. R., Schelhase, M., & Morgan, J. F. (2012). Eating disorders: clinical features and the role of the general psychiatrist. *Advances in psychiatric treatment*, *18*(1), 34-43. http://apt.rcpsych.org/content/18/1/34
- Magarinos et al, Epidemiology and Treatment of Hypochondriasis, CNS Drugs, 2002; 16(1): 9-22
- Morris, P & Lloyd, C (2004) Vocational rehabilitation in psychiatry: a re-evaluation Australian and New Zealand Journal of Psychiatry 38 (7) , 490–494
- NICE (www.nice.org.uk/page.aspx?o=CG026quickrefguide)—Quick reference guide to NICE guidelines for post-traumatic stress disorder
- Phillips KA, Dufresne RG, Jr, Wilkel CS, Vittorio CC. Rate of body dysmorphic disorder in dermatology patients. J Am Acad Dermatol. 2000;42:436-41
- Phillips KA, Gunderson CG, Mallya G, McElroy SL, Carter W. A comparison study of body dysmorphic disorder and obsessive-compulsive disorder. J Clin Psychiatry. 1998;59:568-75.
- Ross CA, Epidemiology of multiple personality disorder and dissociation. Psychiatr Clin North Am 1991, Sept; 14(3): 503-17
- Schneier, FR. Social anxiety disorder. BMJ 2003;327:515-516
- Seivewright, H & Tyrer, P. Change in personality status in neurotic disorders. The Lancet 2002. 359, 9325;2253-2254.
- Soomro, GM et al. Selective serotonin re-uptake inhibitors (SSRIs) versus placebo for obsessive compulsive disorder (OCD).Cochrane Database Syst Rev. 2008 23;(1): CD001765.
- Sumathipala, A. Psychosomatic Medicine 69:889-900 (2007)
- Taylor, CB. Panic disorder. BMJ 2006;332:951-955
- Tyrer, P. & Bateman, A. W. (2004) Drug treatments for personality disorder. Advances in Psychiatric Treatment, 10, 389–398.
- Vedat S, Epidemiology of Dissociative Disorders: An Overview, Epidem Research Int 2011
- Weisberg, R. B. (2009). Overview of generalized anxiety disorder: epidemiology, presentation, and course. *Journal of clinical psychiatry*.
- Wilson, GT & Shafran, R (2005). Eating disorders guidelines from the NICE. The Lancet. 365, 9453;79-81.
- Yakeley, J., & Wood, H. (2014). Paraphilias and paraphilic disorders: diagnosis, assessment and management. *Advances in Psychiatric Treatment*, *20*(3), 202-213.
- Zimmerman, M et al (2005) The Prevalence of DSM-IV Personality Disorders in Psychiatric Outpatients. Am J Psychiatry 162:1911-1918